

# PRODUCT REQUIREMENT DOCUMENT (PRD)

**UMHackathon 2026**

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**Table of Contents**

|                                             |   |
|---------------------------------------------|---|
| 1. Project Overview                         | 3 |
| 2. Background & Business Objective          | 3 |
| 3. Product Purpose                          | 4 |
| 4. System Functionalities                   | 4 |
| 5. User Stories & Use Cases                 | 7 |
| 6. Features Included (Scope Definition)     | 8 |
| 7. Features Not Included (Scope Control)    | 8 |
| 8. Assumptions & Constraints                | 8 |
| 9. Risks & Questions Throughout Development | 8 |

**Project Name:** KITA KOD

**Version:** 1.0

## 1. Project Overview

- **Problem Statement:** SME owners, particularly mini-markets and local retailers, struggle to make timely and informed decisions due to fragmented data and reliance on manual intuition. This leads to overstocking (capital waste) or stockouts (lost revenue), especially during volatile periods like celebration seasons and changing weather patterns.
- **Target Domain:** Domain 2: AI for Economic Empowerment & Decision Intelligence
- **Proposed Solution Summary:** KITA KOD is an AI-powered decision intelligence platform that centralizes internal SME data (sales, inventory) and external signals (Google Trends, weather, holidays). It leverages Z.AI GLM to perform context-aware reasoning, generating accurate demand forecasts and concrete action checklists (e.g., "Order 50kg flour by Tuesday") to optimize economic outcomes.

## 2. Background & Business Objective

- **Background of the Problem:** For decades, SME decision-making has been a manual, linear process based on "gut feeling" and basic spreadsheets. While they possess historical sales data, they lack the analytical capabilities to correlate it with complex external variables like search trends or upcoming school holidays.
- **Importance of Solving This Issue:** SMEs are the backbone of the economy. Accelerating their decision-making process from hours of manual analysis to seconds of AI-driven insights allows them to reduce waste, increase revenue, and remain competitive against larger, data-rich corporations.

- **Strategic Fit / Impact:** KITA KOD aligns with the AI-Native development lifecycle by integrating Z.AI GLM as a core reasoning engine. It empowers SMEs with enterprise-level intelligence, directly contributing to economic empowerment and decision intelligence.

### 3. Product Purpose

**3.1. Main Goal of the System:** To provide an AI-enabled decision partner that transforms raw data into measurable economic improvements (reduced costs, increased revenue) through predictive modeling and optimization.

### 3.2. Intended Users (Target Audience):

| User Type         | Description                               | Use Case                                                       |
|-------------------|-------------------------------------------|----------------------------------------------------------------|
| Mini-Market Owner | Operates small to medium retail shops.    | Predicts weekend demand for perishables based on weather.      |
| Bakery/Caterer    | Manages high-fluctuation seasonal orders. | Optimizes ingredient ordering for festive seasons (Raya, CNY). |
| SME Manager       | Oversees daily operations and staff.      | Adjusts staffing levels based on predicted foot traffic peaks. |

### 3.3. Key Benefits:

- **Waste Reduction:** Precision inventory ordering reduces spoilage.
- **Revenue Growth:** Identifying trending items early to capture market demand.
- **Time Efficiency:** Automates data correlation, saving hours of planning weekly.

## 4. System Functionalities

**4.1. Description:** KITA KOD operates as a Decision Intelligence System through a multi-tier data ingestion mechanism. Using Z.AI GLM, the intelligence layer captures non-obvious relationships between internal sales patterns and external market signals, processing them into a dashboard of visualizations, forecasts, and actionable tasks.

### 4.2. Key Functionalities:

| Functionality          | Description                                                                  | Priority |
|------------------------|------------------------------------------------------------------------------|----------|
| Multi-Tier Data Input  | Supports CSV uploads and manual entry for Sales, Expenses, and Inventory.    | High     |
| External Data Fetch    | Automatically correlates Public Holidays, Weather, and Google Trends.        | High     |
| Demand Forecasting     | Generates 7-day and 30-day sales predictions with range estimates.           | High     |
| Action Checklist       | Converts predictions into concrete tasks (Buy, Hire, Promote).               | High     |
| AI "What-If" Assistant | Allows users to simulate scenarios (e.g., "What if I raise prices by 10%?"). | Medium   |

### 4.3. AI Model & Prompt Design

**4.3.1. Model Selection:** Z.AI GLM (General Language Model). Z.AI GLM provides superior contextual reasoning and the ability to process unstructured external data alongside structured internal metrics. Its explainability features ensure SME owners understand the "why" behind every recommendation.

#### 4.3.2. Prompting Strategy:

| Strategy           | Description                                                                    | Justification                                                             |
|--------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Multi-step Agentic | System first cleans data, then correlates with external signals, then reasons. | Necessary for complex multi-variable analysis (weather + trends + sales). |
| Few-shot Prompting | Providing examples of SME data and expected insight formats.                   | Ensures high-quality, actionable, and consistent recommendations.         |

**Approach:** We use a Chain-of-Thought prompting approach where the GLM is instructed to first identify patterns, then hypothesize impacts of external variables, and finally formulate specific actions.

#### 4.3.3. Context & Input Handling:

| Parameter          | Value                                                             |
|--------------------|-------------------------------------------------------------------|
| Maximum Input Size | 8,000 tokens                                                      |
| Handling Method    | Chunking & Summarization                                          |
| Overflow Behaviour | Oldest sales data is summarized into trends before being dropped. |

#### Input Size Limits:

- Text: 50,000 characters
- File Upload: 5MB (CSV)
- API Request: 2MB

#### 4.3.4. Fallback & Failure Behavior:

| Scenario           | Behaviour                                                                 |
|--------------------|---------------------------------------------------------------------------|
| Off-topic Response | System detects low confidence and asks for specific data points.          |
| Unusable Output    | Automatic retry with a more restrictive system prompt.                    |
| API Timeout        | 30s timeout with fallback to basic statistical analysis (no AI insights). |
| Rate Limiting      | Local queuing system to throttle requests to Z.AI GLM.                    |
| Model Unavailable  | Displays "Maintenance Mode" with cached previous results.                 |

## 5. User Stories & Use Cases

- User Stories:**

| ID     | User Story                                                                                       | Priority |
|--------|--------------------------------------------------------------------------------------------------|----------|
| US-001 | As a mini-market owner, I want to see a 7-day forecast so I can order the right amount of bread. | High     |
| US-002 | As a baker, I want to know if "Pineapple Tarts" are trending so I can start production early.    | High     |
| US-003 | As a manager, I want an action checklist so I don't forget to hire extra staff for the holidays. | High     |

- Use Cases (Main Interactions):**

| Use Case ID | Description              | Actor | Pre-condition       | Post-condition           | Steps                                                                    |
|-------------|--------------------------|-------|---------------------|--------------------------|--------------------------------------------------------------------------|
| UC-001      | Generate Demand Forecast | Owner | Sales data uploaded | Forecast chart displayed | 1. Navigate to Predictions<br>2. Select timeframe<br>3. Click 'Generate' |
| UC-002      | Scenario Simulation      | Owner | System active       | Scenario impact report   | 1. Open AI Assistant<br>2. Enter "What if price +10%"                    |

| Use Case ID | Description | Actor | Pre-condition | Post-condition | Steps           |
|-------------|-------------|-------|---------------|----------------|-----------------|
|             |             |       |               |                | 3. View analyst |

## 6. Features Included (Scope Definition)

| Feature                   | Description                                    | Deliverable    | Priority |
|---------------------------|------------------------------------------------|----------------|----------|
| Sales Dashboard           | Visualizes historical trends and margins.      | Interactive UI | High     |
| External Data Integration | Real-time fetch of weather and holidays.       | Data Pipeline  | High     |
| GLM Insights              | Plain-language explanations for all forecasts. | AI Module      | High     |

## 7. Features Not Included (Scope Control)

| Feature              | Reason for Exclusion           | Phase   |
|----------------------|--------------------------------|---------|
| POS Integration      | Complex hardware dependencies. | Phase 2 |
| Automatic Purchasing | Requires supplier API access.  | Phase 3 |

## 8. Assumptions & Constraints

| Type       | Description                                                          |
|------------|----------------------------------------------------------------------|
| Assumption | Users have access to basic CSV sales exports from their POS systems. |
| Constraint | Response time for AI insights must be under 10 seconds.              |
| Constraint | Must be mobile-responsive for on-the-floor use.                      |

## 9. Risks & Questions Throughout Development

| Risk/Question     | Impact | Mitigation                                         | Status    |
|-------------------|--------|----------------------------------------------------|-----------|
| Data Privacy      | High   | All SME data is anonymized before reaching GLM.    | Mitigated |
| GLM Hallucination | Medium | Validating AI actions against hard business rules. | Open      |